

Short Proof Sketch: DP implies Stability



Stability (small change in dataset leads to small change in output distribution) $\Rightarrow$ generalization. DP provides distributional stability; hence private algorithms often generalize well.

Practical Considerations



- ▶ Choosing ε : policy decision; common values: 0.01–1 for strong privacy, up to 8 for weaker.
- ▶ Accounting: composition bookkeeping.
- ▶ Implementation: use careful numerical methods; avoid releasing intermediate non-private outputs.



- ▶ DP provides strong, composable privacy guarantees.
- ▶ Laplace and Gaussian mechanisms are workhorses.
- ▶ Advanced mechanisms (exp mech, objective perturbation) enable private learning.
- ▶ Utility depends on sensitivity, ϵ , and sample size.

Motivation



Labels are expensive; unlabeled data is cheap. Semi-supervised learning (SSL) uses both to improve performance.

Have a labeled set $S_l = \{(x_i, y_i)\}_{i=1}^n$ and an unlabeled set $S_u = \{x_j\}_{j=1}^m$. Goal: learn function $h: \mathcal{X} \rightarrow \mathcal{Y}$ with lower error than using labeled data alone.

Common Assumptions



- ▶ **Cluster assumption:** decision boundary passes through low density regions.
- ▶ **Manifold assumption:** data lies on a low-dimensional manifold.
- ▶ **Smoothness:** nearby points should have similar labels.

Taxonomy of Methods



- ▶ Self-training / bootstrapping
- ▶ Co-training
- ▶ Graph-based methods (label propagation)
- ▶ Generative models (EM-style)
- ▶ Consistency-regularization in deep learning

Worked Example: Self-Training (Toy)



Labeled: two points $(0,0) \rightarrow 0$, $(1,1) \rightarrow 1$. Unlabeled: $(0.1, 0.1)$, $(0.9, 0.9)$. Train classifier on labeled points, predict on unlabeled, add high-confidence predictions as labels, retrain.

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Graph-Based SSL: Label Propagation



Construct graph $G = (V, E)$ over all points; edge weights w_{ij} reflect similarity. Solve harmonic function: minimize $\sum_{i,j} w_{ij}(f(i) - f(j))^2$ subject to labeled values fixed.

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Construct graph $G = (V, E)$ over all points; edge weights w_{ij} reflect similarity. Solve harmonic function: minimize $\sum_{i,j} w_{ij}(f(i) - f(j))^2$ subject to labeled values fixed. Solution: set of linear equations solved by Laplacian system.

Worked Example: Small Graph



Three unlabeled nodes A,B,C connected chain: A–B–C. Labels:
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Three unlabeled nodes A,B,C connected chain: A–B–C. Labels: $A=0$, $C=1$. Harmonic solution gives $B=0.5$. This agrees with smoothness assumption.

Transductive vs Inductive Learning



- ▶ Transductive: predict labels only for given unlabeled set S_u .
- ▶ Inductive: learn function to generalize to unseen data.

Sample Complexity: Intuition



Unlabeled data helps by reducing uncertainty about data distribution (e.g., cluster boundaries). Formalizing this yields reductions in label complexity under assumptions.

PAC-style Result (Informal)



Under cluster assumption, if clusters are well-separated, learning each cluster's label may require only $O(\log(1/\delta)/\alpha)$ labeled examples instead of $O(1/\alpha^2)$.

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Under cluster assumption, if clusters are well-separated, learning each cluster's label may require only $O(\log(1/\delta)/\alpha)$ labeled examples instead of $O(1/\alpha^2)$. Therefore, unlabeled data can exponentially reduce label complexity in benign cases.

Rademacher Complexity in SSL



One approach: use unlabeled data to reduce hypothesis class complexity via data-dependent constraints (e.g., margin on manifold). Rademacher bound can include terms depending on unlabeled data geometry.

Short Proof: Unlabeled Data Reduces Hypothesis Space



Sketch: If unlabeled data reveals structure that rules out hypotheses inconsistent with clustering, effective hypothesis class size $|\mathcal{H}_{\text{eff}}|$ shrinks, reducing sample complexity $O(\log |\mathcal{H}_{\text{eff}}|/\alpha^2)$.

Transductive SVM (Idea)



Find decision boundary that maximizes margin on both labeled and unlabeled data, with unlabeled points treated as having labels to be chosen to maximize margin.

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Find decision boundary that maximizes margin on both labeled and unlabeled data, with unlabeled points treated as having labels to be chosen to maximize margin. Non-convex, but yields improved boundaries in practice.

Graph Laplacian Regularization



Minimize

$$\sum_{i=1}^n \ell(h(x_i), y_i) + \lambda \sum_{i,j} w_{ij} (h(x_i) - h(x_j))^2.$$

Second term encourages smoothness w.r.t. graph geometry.

Worked Example: Harmonic Function on a Small Graph



Consider 4 nodes: 1 and 4 labeled (0 and 1), nodes 2 and 3 unlabeled. Chain edges with weight 1. Solve linear system for nodes 2 and 3: yields values $1/3$ and $2/3$ respectively.

Consistency Regularization (Deep SSL)



Methods like "Mean Teacher", VAT, FixMatch encourage model to give consistent outputs under input perturbations; use unlabeled loss term.

Worked Example: Pseudo-Labeling



Train on labeled data, predict labels for unlabeled, keep high-confidence predictions as labels, then retrain. Simple, often effective.

Limitations and Pitfalls



- ▶ Wrong assumptions (e.g., clusters don't align with labels) can hurt.
- ▶ Self-training can amplify errors.
- ▶ Need careful validation and calibration.

Combining DP and SSL



Interesting research area: use DP when labels are sensitive; or use unlabeled public data with private labeled data. Challenges: privacy budget, unlabeled distribution shift.

Practical Tips



- ▶ Start with simple baselines (self-training, label propagation).
- ▶ Validate assumptions: visualize clusters, manifold structure.
- ▶ For DP: track privacy budget and calibrate ϵ to risk tolerance.



A short list to get started:

- ▶ Dwork & Roth, "The Algorithmic Foundations of Differential Privacy" (2014).
- ▶ Chapelle, Scholkopf, Zien (eds.), "Semi-Supervised Learning" (2006).
- ▶ Zhu, Ghahramani, Lafferty: "Semi-Supervised Learning Using Gaussian Fields and Harmonic Functions" (2003).
- ▶ Kasiviswanathan et al., "What Can We Learn Privately?" (2008).